

AWS EMR & Hadoop Overview

A simple guide to understanding Hadoop fundamentals and how Amazon EMR simplifies big data processing in the cloud.

AWS Machine Learning Associate - Part 6

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The Challenge of Big Data

Single Computer Limits

When dataset size grows into terabytes or petabytes, a single computer runs out of memory (RAM), disk storage space, and CPU capacity required to process queries in reasonable time.

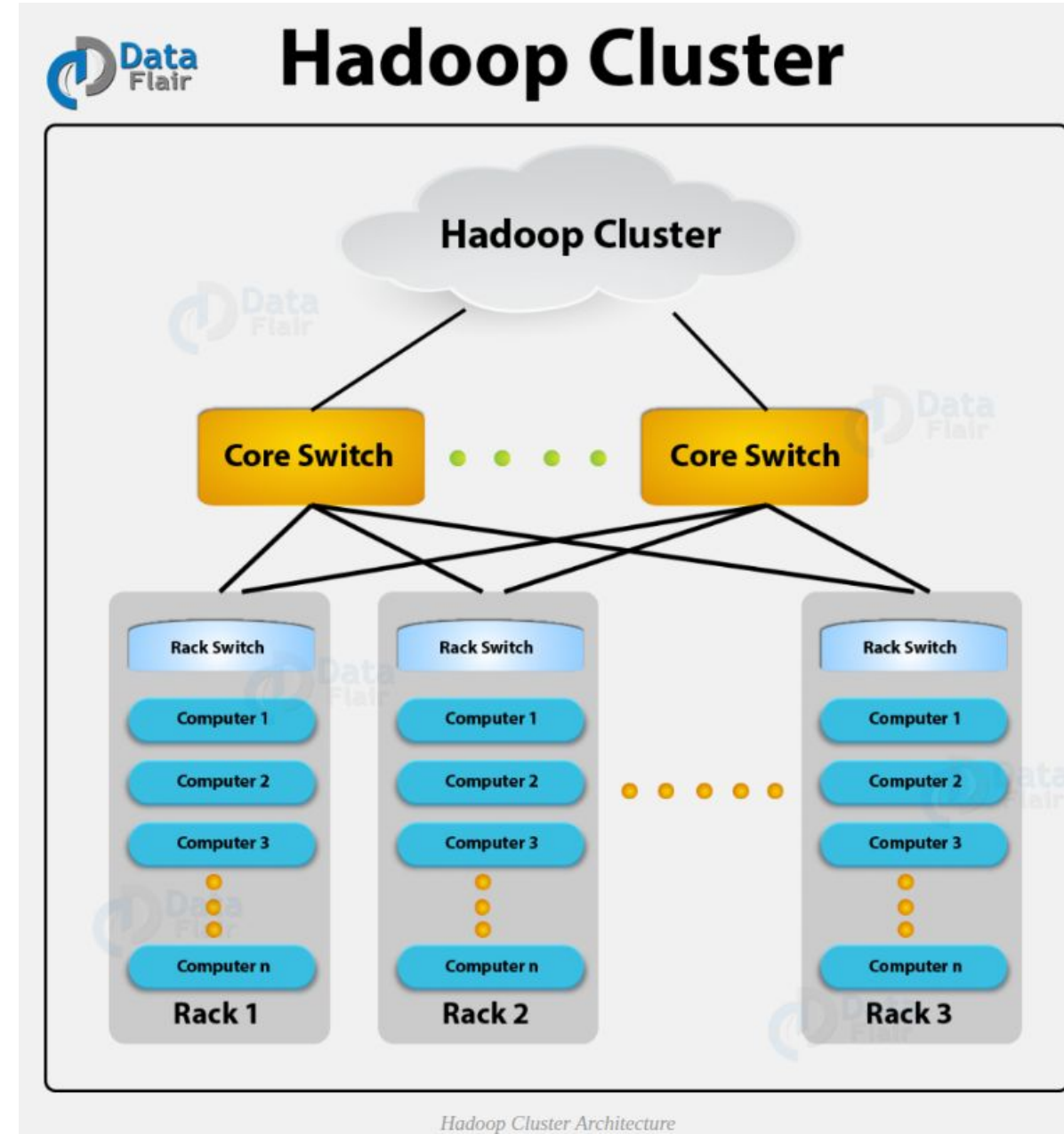
Distributed Solution

Instead of using one large machine, we connect hundreds of smaller commodity computers into a **Cluster**. They share the workload by storing and processing parts of data in parallel.

Apache Hadoop Core Architecture

The Open-Source Big Data Standard

-  **HDFS (Storage):** Hadoop Distributed File System splits large files into blocks and spreads them across nodes.
-  **YARN (Resource Manager):** Manages cluster computing resources and schedules jobs.
-  **MapReduce (Processing Engine):** Software framework that processes huge datasets in parallel.

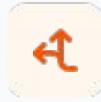


How MapReduce Works Simply



1. Map Phase

Breaks incoming raw data into smaller key-value pairs across worker nodes (e.g., counting word occurrences across thousands of documents).



2. Shuffle & Sort

Gathers and organizes all matching key-value pairs from different nodes so identical keys are grouped together.



3. Reduce Phase

Combines and aggregates the sorted data (e.g., calculating final sums or averages) to produce the unified answer.

Amazon EMR Cloud Big Data

Managed Hadoop in AWS

Managed Service: Amazon EMR (Elastic MapReduce) automates provisioning, configuring, and tuning Hadoop clusters on AWS.

Ecosystem Support: Runs Apache Spark, Hive, HBase, Presto, and Flink without manual setup.

S3 Integration (EMRFS): Decouples compute from storage, letting you store data cheaply in Amazon S3 while running clusters on demand.



Anatomy of an EMR Cluster



Primary (Master) Node

Coordinates the cluster, manages YARN resources, tracks job status, and distributes tasks to worker nodes.



Core Nodes

Runs processing tasks AND stores data locally using HDFS. Always stays running during cluster operations.



Task Nodes

Runs processing tasks ONLY (no persistent storage). Perfect for auto-scaling during workload spikes using cheap EC2 Spot Instances.

On-Prem Hadoop vs Amazon EMR

Comparison Metric	Traditional On-Premises Hadoop	Amazon EMR (Cloud Managed)
Setup & Provisioning	Months of procurement and manual software tuning	Minutes via AWS Console / API
Scaling Flexibility	Hard to scale; fixed physical hardware capacity	Instant Auto-Scaling up & down
Storage Architecture	Coupled (Storage & Compute on same servers)	Decoupled (Store in S3, compute in EMR)
Cost Model	High upfront CapEx for physical servers	Pay-as-you-go per second

Key Takeaways



1. Hadoop introduced distributed computing



2. Amazon EMR made it scalable



3. Cost-efficient big data clusters on demand



4. Effortless in the cloud.